ZENTRAL HACK 2026 · AGENT HARNESS

Spec in.
Strategy out.



An agent harness for games it has never seen.

Elias · Isaak · Timo

THE TASK · THE INSIGHT

An unseen game, a fixed budget, only wins count

The task

A $0.03 · 3 min

B $0.50 · 10 min

C $10 · 30 min

What research says

85% vs **30%**

LLM writes code vs. LLM plays

Code World Models, Google DeepMind

Our insight

The LLM is the programmer, not the player.

Sources: Zentral Hack 2026 task description; Code World Models for General Game Playing, arXiv 2510.04542.

THE HARNESS

Many drafts, one verified simulator, one champion

spec.md in

1
Write
drafts + a simulator

2
Verify
against the spec

3
Self-play
real tournament rules

4
Improve
challengers vs. champion

5
Guard
never a forfeit

strategy.py out

↻ repeat while time and money last

HOW WE BUILT IT

Built like it works: measure, decide, repeat

8

games, 2 held out

58

measured runs

≈ $21

API cost

Built with Claude Code · 216 automated tests

EPOCH 1

Ablation

kept only what wins

EPOCH 2

Tournament conditions

selection = reality

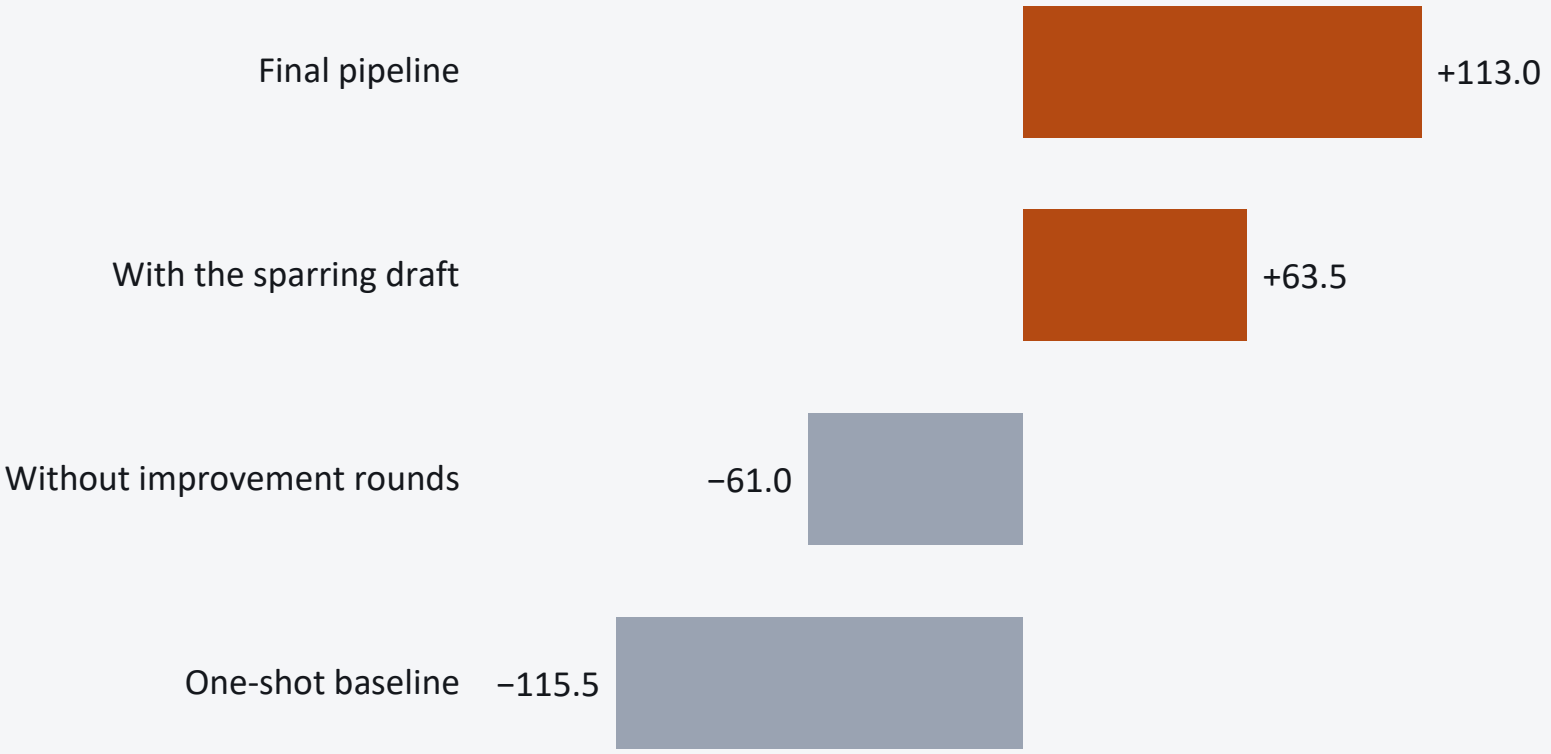
EPOCH 3

Failure mining

3 defects fixed

EPOCH 1 · RESULT

Every feature had to earn its place



48–0–0

wins – draws – losses against the one-shot baseline

Net points, 3 games, +1 / 0 / −1 per match

THE EXAM

It transfers to games it was never tuned on

Wins – draws – losses against the one-shot baseline, on 2 games never used for tuning

44–1–3

held-out game 1

32–0–16

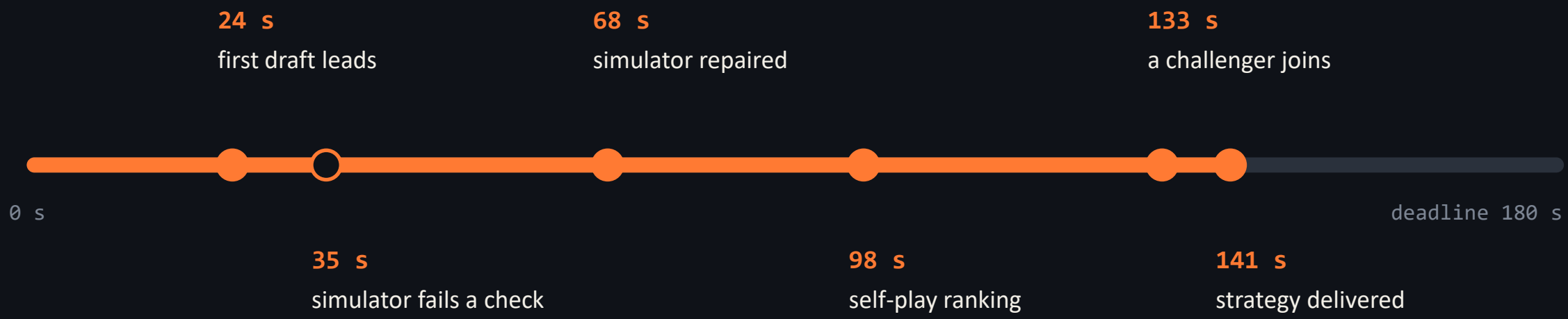
held-out game 2

53 / 53

strategies passed, zero forfeits

DEMO · ONE REAL RUN, FROM SPEC TO STRATEGY.PY

141 seconds, 1.8 cents, one champion



Spec in. Strategy out. Never a forfeit.